MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION NOVEMBER 2025

FIRST SEMESTER

MULTI-DISCIPLINARY COURSES (MDC) – MG1MDCGEO100 – UNDERSTAND THE EARTH

(2024 ADMISSION ONWARDS)

Duration: 1 Hour **Maximum Marks: 35**

ANSWER KEY

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short notes in 60 words
Answer any **7** questions
Each question carries **2** marks

1. State Big Bang theory. [K] / [CO1]

Big Bang theory attributes the creation of the universe to 'very rapid, instantaneous, exponential expansion of an infinitely small, very hot and infinitely dense phase (Definition- 2 marks).

2. Define earthquake. [K] / [CO3]

An earthquake is the sudden shaking or trembling of the ground caused by the sudden release of energy stored in rocks beneath the Earth’s surface. This energy travels outward as seismic waves, making the ground vibrate (Definition- 2 marks)

3. State the difference between intrusive and extrusive igneous rock. [U] / [CO2]

Intrusive igneous rocks form when magma cools slowly beneath the Earth's surface (Definition- 1 mark). Extrusive igneous rocks form when lava cools quickly on or above the Earth's surface, often after a volcanic eruption (Definition- 1 mark).

4. Describe about short impacts of landslides. [U] / [CO4]

Immediate loss and damage: Landslides cause sudden loss of life, injuries, and destruction of houses, roads, and infrastructure due to the rapid movement of soil, rock, or debris (1 mark). Environmental and service disruption: They result in soil erosion, loss of vegetation, blockage of rivers, and interruption of transport and essential services (1 mark).

5. Identify any two ore minerals of iron. [K] / [CO3]

Hematite, Magnetite, Goethite, Limonite, Siderite (Any two-1 mark each)

6. Define porosity and permeability. [K] / [CO3]

Porosity is the percentage of empty spaces found between the grains of soil sediment or rock (Definition- 1 mark). Permeability is the ability of a material to allow water to flow through it (Definition- 1 mark).

7. Describe the role of Engineering Geology in construction projects. [U] / [CO2]

Engineering Geology applies geological data and principles to the design and construction of large engineering projects, such as dams, tunnels, bridges, and buildings. It ensures that the geological conditions of a site are suitable and safe for construction (Application- 2 marks).

8. Describe divergent plate boundary. [U] / [CO3]

Divergent plate boundaries are regions where two tectonic plates move away from each other (Definition with figure- 2 marks).

Part B

Short notes in 230 words
Answer any 3 questions
Each question carries 7 marks

9. Describe disaster mitigation and preparedness. [U] / [CO4]

Definition of disaster mitigation and preparedness (2 marks); Mitigation measures like safe construction practices, strengthening infrastructure (2 marks); Preparedness activities like developing emergency plans and procedures, conducting training and drills (2 marks); Importance of mitigation and preparedness (1 mark).

10. Describe fossil fuel, list the types of fossil fuels and explain the process of its formation. [U] / [CO3]

Petroleum, coal, and natural gas are called fossil fuels- they formed from the remains of plants and animals (Definition of fossil fuel-1 mark)

Process of coal formation - coalification in detail (3 marks).

Crude oil (Petroleum) and natural gas formation (3 marks).

11. What are the major exogenous processes? [U] / [CO3]

Definition of exogenous process (1 mark); Weathering, types of weathering (3 marks); erosion, transportation, deposition (3 marks).

12. Describe how the lithosphere, biosphere, and hydrosphere interact to support life. [U] / [CO1]

Definition of lithosphere, biosphere, hydrosphere (1 mark)

Lithosphere–Biosphere interaction: The lithosphere provides soil, minerals, and nutrients that support plant growth (producers). Ecosystem formation: Plants, animals, and decomposers in the

biosphere depend on lithospheric soil and environment to form ecosystems such as forests, deserts, ponds, and marine systems (2 marks).

Hydrosphere–Biosphere interaction: The hydrosphere provides water required for photosynthesis, respiration, and survival of all living organisms. Aquatic ecosystems support diverse life forms. Role of the water cycle (2 marks)

Lithosphere–Hydrosphere interaction: Water infiltrates the lithosphere to form groundwater; rock weathering by water releases nutrients that plants absorb (1 mark).

Combined effect supporting life: Nutrients from lithosphere + water from hydrosphere + living organisms in biosphere together maintain food chains, nutrient cycles, and balanced ecosystems necessary for life (1 mark)

13. Identify impact of landslides. [K] / [CO4]

Definition of landslides (1 mark); Loss of life and injuries, Damage to property and infrastructure, Damage to agriculture and forests, River blockage and flooding, Economic loss and displacement of communities (Any four impacts- 1.5 marks each)

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